Nutrition

Mark scheme and worked solutions for 20 questions: 7 multiple choice, 13 structured.

---

1 · Part 2

Paper 1 & 2 Nutrition [1]

Answer: B

Explanation

Iron is needed to make haemoglobin, the protein in red blood cells that carries oxygen. Calcium is needed for bones and teeth, vitamin D helps the body absorb calcium, and dietary fibre adds bulk to food to help it move through the gut; none of these is used to make haemoglobin.

---

2 · Part 2

Paper 1 & 2 Nutrition [1]

Answer: D

Explanation

Light enters through the upper surface, so the palisade layer just below the upper epidermis receives the most light for its chloroplasts to absorb. It is the closest layer to the light, not the furthest (A); its position is not chosen because it is warmer (B); and palisade cells, like all plant cells, have cellulose cell walls (C).

---

3 · Part 2

Paper 1 & 2 Nutrition [1]

Answer: A

Explanation

Amylase digests starch into smaller sugar molecules, and maltase then digests these into glucose. B has the order reversed. Lipase digests lipids, not sugars (C), and protease digests proteins, not starch (D).

---

4 · Part 2

Paper 1 & 2 Nutrition [1]

Answer: C

Explanation

Food leaving the stomach is acidic. Alkaline bile neutralises this acid so that the pH in the small intestine is suitable for its enzymes. Bile is not an enzyme and does not chemically digest lipids (A); it does not kill all bacteria (B); and being alkaline, it makes the contents less acidic, not more acidic (D).

Topical Hub
topical-hub.pages.dev

5 · Part 2 Paper 1 & 2 Nutrition [1]

Answer: B

Explanation

Peristalsis is waves of contraction of the muscles in the wall of the gut, which squeeze food along. The muscles do not all contract at once (A); peristalsis happens along the whole gut, not only the stomach, and does not depend on gravity (C); villi absorb digested food and do not push food along (D).

6 · Part 2 Paper 1 & 2 Nutrition [1]

Answer: A

Worked solution

Always divide the change by the original (starting) value. 3150 ÷ 10 500 = 0.30, and 0.30 × 100 = 30%.

7 · Part 2 Paper 2 only Nutrition [1]

Answer: C

Explanation

The volume (mass) of water must be the same for every food, because the same energy would cause a larger temperature rise in a smaller volume of water. The type of food is the independent variable (A), the temperature rise is the dependent variable that is measured (B), and the energy content is what is being worked out (D).

Topical Hub
topicalhubpages.dev

8 · Part 2

Paper 1 & 2 | Nutrition | [8]

Mark scheme

(a) **[2]**

- carbon dioxide + water (as reactants, on the left) (1)
- → glucose + oxygen (as products, on the right) (1)

IGNORE light or chlorophyll written above the arrow

ALLOW sugar for glucose

(b) **[2]**

- correct formulae: $CO_2 + H_2O \rightarrow C_6H_{12}O_6 + O_2$ (1)
- correctly balanced: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (1)

ALLOW formulae written without subscripts, e.g. CO2, C6H12O6

(c) **[1]**

- chlorophyll (1)

(d) **[3]**

- light energy is converted into chemical energy (1)
- the chemical energy is stored in glucose (1)
- the glucose is used in respiration to release energy for the plant OR is converted to starch as an energy store OR is used to make other substances such as cellulose for growth (1)

Worked solution

(a) The two raw materials are carbon dioxide (from the air) and water (from the soil). The products are glucose and oxygen: carbon dioxide + water → glucose + oxygen. Light energy is absorbed by chlorophyll to drive the reaction.

(b) The balanced symbol equation is $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. Check the balancing: 6 carbon atoms, 12 hydrogen atoms and 18 oxygen atoms on each side.

(c) The green pigment is chlorophyll, found in chloroplasts.

(d) Photosynthesis converts light energy into chemical energy, stored in glucose. This matters because the plant can then use the glucose in respiration to release energy for its cells, or store it as starch, or use it to build new substances for growth. Without this conversion, the plant could not use the energy in light.

Topical Hub
topical-hub.pages.dev

9 · Part 2

Paper 1 & 2 Nutrition [8]

Mark scheme

(a) **[2]**

- rate increases as light intensity increases from 0 to 40 arbitrary units (1)
- rate then stays constant (levels off) at 30 arbitrary units between 40 and 50 arbitrary units (1)
- the second mark needs a data quote, e.g. stays at 30 OR levels off at 40 arbitrary units

ALLOW rate increases less steeply as light intensity increases

(b) **[2]**

- light intensity is no longer the limiting factor (1)
- carbon dioxide concentration is the limiting factor OR the carbon dioxide concentration is too low for the rate to increase further (1)

(c) **[2]**

- at low light intensity, light intensity is the limiting factor (1)
- so increasing the carbon dioxide concentration has no effect on the rate OR carbon dioxide is not limiting at this light intensity (1)

(d) **[2]**

- (48 − 30) ÷ 30 × 100 OR 18 ÷ 30 × 100 (1)
- 60 (%) (1)

REJECT 37.5% (dividing by 48)

Worked solution

(a) At 0.04% carbon dioxide, the rate rises from 0 to 30 arbitrary units as light intensity increases from 0 to 40 arbitrary units, then stays at 30 between 40 and 50 arbitrary units.

(b) A limiting factor is the factor in shortest supply, which stops the rate going any higher. Above 40 arbitrary units, more light makes no difference, so light is no longer limiting. Something else is: here the carbon dioxide concentration. We know this because, at the same light intensity, the plants with more carbon dioxide (0.10%) photosynthesise faster.

(c) At very low light intensity, light is the limiting factor. Giving the plants more carbon dioxide cannot speed up photosynthesis because there is not enough light, so both chambers have the same rate.

(d) Change = 48 − 30 = 18. Percentage increase = (18 ÷ 30) × 100 = 60%. Divide by the original value (30), not the new value.

Topical Hub
topical-hub.pages.dev

10 · Part 2

Paper 1 & 2 Nutrition [7]

Mark scheme

(a) **[2]**

- volume of gas increases as temperature increases from 10 °C to 30 °C, reaching a maximum of 34 mm^3 at 30 °C (1)
- volume of gas then decreases as temperature increases above 30 °C, to 2 mm^3 at 50 °C (1)
- at least one data quote with units needed for full marks

(b) **[2]**

- molecules (enzymes and substrates) have more kinetic energy OR move faster (1)
- so there are more collisions between enzymes and substrate molecules OR more enzyme–substrate complexes form, so the reactions of photosynthesis are faster (1)

(c) **[2]**

- enzymes (that control photosynthesis) are denatured (1)
- the shape of the active site changes, so the substrate no longer fits (1)

REJECT enzymes are killed

(d) **[1]**

- repeat the investigation at smaller temperature intervals between 20 °C and 40 °C, e.g. every 2 °C (1)

ALLOW any range that includes 30 °C with smaller intervals, e.g. 25 °C to 35 °C

Worked solution

(a) The volume of gas rises from 8 mm^3 at 10 °C to a peak of 34 mm^3 at 30 °C, then falls to only 2 mm^3 at 50 °C.

(b) Photosynthesis is controlled by enzymes. As temperature rises, the enzyme and substrate molecules have more kinetic energy and collide more often, so more enzyme–substrate complexes form and the rate increases.

(c) Above the optimum temperature, the enzymes start to denature. The active site changes shape, so substrate molecules no longer fit, and the rate falls sharply.

(d) The results only show that the optimum is somewhere between 20 °C and 40 °C. Testing more temperatures in this range, closer together, would show the optimum more precisely.

Topical Hub
topical-hub.pages.dev

11 · Part 2 Paper 1 & 2 Nutrition [6]

Mark scheme

(a)

(a)(i) **[1]**

- large surface area to absorb (more) light (1)

(a)(ii) **[1]**

- light can reach all the cells (of the mesophyll) OR short distance for carbon dioxide to diffuse to the cells (1)

(a)(iii) **[1]**

- lets light pass through to the palisade (mesophyll) cells (1)

ALLOW lets light reach the chloroplasts

(a)(iv) **[1]**

- xylem supplies water (a raw material) for photosynthesis (1)

IGNORE xylem supports the leaf

(b) **[2]**

- light intensity is low in shade OR light is the limiting factor (1)
- larger leaves have a larger surface area to absorb more of the light available, so more photosynthesis OR more glucose is made (1)

Worked solution

(a) (i) A broad, flat leaf has a large surface area, so it can absorb more light. (ii) A thin leaf means light can reach all the photosynthesising cells, and carbon dioxide only has a short distance to diffuse. (iii) The upper epidermis cells are transparent, so light passes straight through them to the palisade cells below, which contain many chloroplasts. (iv) Xylem in the veins brings water to the leaf; water is one of the raw materials for photosynthesis.

(b) In deep shade, light intensity is low, so light limits the rate of photosynthesis. A larger leaf absorbs more of the little light there is, so the plant can still make enough glucose to grow.

Topical Hub
topical-hub.pages.dev

12 · Part 2 Paper 1 & 2 Nutrition [8]

Mark scheme

(a) **[2]**

- (12.5 − 4.0) ÷ 12.5 × 100 OR 8.5 ÷ 12.5 × 100 (1)
- 68 (%) (1)

REJECT 212.5% (dividing by 4.0)

(b) **[3]**

- nitrate ions are needed to make amino acids (1)
- amino acids are needed to make proteins (1)
- proteins are needed for growth OR to make new cells OR to make enzymes, so the plant grows less (1)

(c) **[3]**

- magnesium ions are needed to make chlorophyll (1)
- less chlorophyll, so less light (energy) is absorbed (1)
- so less photosynthesis, so less glucose is made for growth OR for respiration OR to make new substances such as cellulose and starch (1)

ALLOW yellow leaves show there is less chlorophyll

Worked solution

(a) Decrease = 12.5 − 4.0 = 8.5 g. Percentage decrease = (8.5 ÷ 12.5) × 100 = 68%. Always divide by the original value (the complete solution).

(b) Plants use nitrate ions to make amino acids, which are joined together to make proteins. Without proteins the plant cannot make new cells or enzymes, so it cannot grow well. This is why these plants are small and have the lowest dry mass.

(c) Magnesium ions are needed to make chlorophyll. Without them the leaves turn yellow because they have less chlorophyll. Less light energy is absorbed, so less photosynthesis takes place and less glucose is made. With less glucose for respiration and for building new substances, the plant grows less.

Topical Hub
topical-hub.pages.dev

13 · Part 2

Paper 1 & 2 Nutrition [7]

Mark scheme

(a) **[2]**

- independent variable: light intensity OR distance of the lamp from the boiling tube (1)
- dependent variable: number of bubbles (of gas) released in 1 minute OR rate of photosynthesis (1)

(b) **[2]**

- so that carbon dioxide is not a limiting factor (1)
- so that only light intensity affects the rate of photosynthesis OR so that carbon dioxide is available for photosynthesis (1)

(c) **[2]**

- the temperature of the water changes, and temperature affects the rate of photosynthesis (so temperature is not controlled) (1)
- place a heat shield, e.g. a beaker or tank of water, between the lamp and the boiling tube OR use an LED lamp that gives out little heat OR monitor the temperature with a thermometer and keep it constant using a water bath (1)

(d) **[1]**

- collect the gas and measure its volume, e.g. in a gas syringe OR an inverted measuring cylinder OR a capillary tube (1)

ALLOW bubbles may be different sizes, so measure the volume of gas

Worked solution

(a) The independent variable is the one Jamal changes: light intensity, by moving the lamp. The dependent variable is the one he measures: the number of bubbles released per minute.

(b) Sodium hydrogencarbonate makes sure there is plenty of carbon dioxide in the water. Then carbon dioxide does not limit photosynthesis, so any change in rate is caused by light intensity.

(c) A lamp close to the tube warms the water, and temperature also affects the rate of photosynthesis. This would make it a second, uncontrolled variable. A beaker of water between the lamp and the tube absorbs the heat, or an LED lamp can be used.

(d) Bubbles can be different sizes, so counting them is not accurate. Collecting the gas and measuring its volume, for example in a gas syringe, is better.

Topical Hub
topical-hub.pages.dev

14 · Part 2

Paper 1 & 2 Nutrition [9]

Mark scheme

(a) **[2]**

- to remove (use up) the starch already in the leaves OR to destarch the plant (1)
- so any starch found after the experiment must have been made during the experiment (1)

(b) **[4]**

Award one mark for each point, up to a maximum of 4.

- place the leaf in boiling water (for about 1 minute) to kill the cells OR to break down the cell membranes OR to stop the reactions (1)
- place the leaf in hot ethanol to remove the chlorophyll OR to decolourise the leaf (1)
- heat the ethanol in a water bath OR turn off the Bunsen burner before using ethanol, because ethanol is flammable (1)
- rinse the leaf in (warm) water to soften it (1)
- add iodine solution (to the leaf on a white tile) (1)
- starch is shown by a blue-black colour (1)

ALLOW wear eye protection as the safety precaution if the ethanol hazard is not given

(c) **[3]**

- covered part: orange-brown OR yellow-brown OR brown (1)
- uncovered part: blue-black (1)
- light is needed for photosynthesis, so the covered part made no glucose (to convert to starch), while the uncovered part made glucose and stored it as starch (1)

REJECT blue for blue-black; REJECT the covered part stays green

Topical Hub
topical-hub-pages.dev

Worked solution

(a) Keeping the plant in the dark for 48 hours means the leaves use up the starch they already contain. Any starch found at the end must have been made during the 24 hours of light.

(b) Boil the leaf in water for about a minute to kill the cells. Put it in hot ethanol to remove the chlorophyll, so the colour change can be seen. Ethanol is flammable, so heat it in a water bath with the Bunsen burner turned off. Rinse the leaf in water to soften it, spread it on a white tile and add iodine solution. Starch turns the iodine solution blue-black.

(c) The covered part had no light, so it could not photosynthesise or make starch; it stays orange-brown (the colour of iodine solution). The uncovered part received light, made glucose by photosynthesis and stored it as starch, so it turns blue-black.

15 · Part 2

Paper 1 & 2 Nutrition [7]

Mark scheme

(a) **[2]**

- cheese (1)
- it contains the most calcium, 720 mg per 100 g (1)
- the reason mark needs calcium; ALLOW the comparison without the number

(b) **[1]**

- needed for healthy skin and gums OR prevents scurvy OR needed for healing wounds (1)

(c) **[2]**

- 25.0 × 60 ÷ 100 OR 25.0 × 0.6 (1)
- 15 g (1)

(d) **[2]**

- cheese contains no dietary fibre (1)
- fibre adds bulk to food so that it can be pushed through the gut by peristalsis OR so faeces can be egested easily (1)

Worked solution

(a) Calcium is needed for strong bones and teeth. Cheese has by far the most calcium (720 mg in 100 g), so cheese is the best choice.

(b) Vitamin C is needed to keep skin and gums healthy; a lack of it causes scurvy.

(c) 100 g of cheese contains 25.0 g of protein, so 60 g contains 25.0 × 60 ÷ 100 = 15 g of protein.

(d) The table shows cheese has 0 g of dietary fibre. Fibre adds bulk to the food in the gut, which helps the gut muscles push it along by peristalsis. Without fibre, food moves slowly through the large intestine and faeces become hard, causing constipation.

Topical Hub
topical-hub.pages.dev

16 · Part 2

Paper 2 only | Nutrition | [8]

Mark scheme

(a) **[2]**

- the girl aged 15 is still growing (rapidly) (1)
- energy is needed to make new cells OR new tissues OR to build proteins (1)

ALLOW has a higher metabolic rate OR more respiration

(b) **[2]**

- energy is needed for the growth of the fetus (and placenta) (1)
- she has extra body mass to carry and move OR her metabolic rate is higher (1)

(c) **[2]**

- energy transferred = 25 × 4.2 × 12 = 1260 J (1)
- 1260 ÷ 0.80 = 1575 OR 1600 J per g (1)

ALLOW 1580 J per g

ECF their energy transferred ÷ 0.80

(d) **[2]**

- extra energy = 9200 − 8400 = 800 kJ (1)
- 800 ÷ 16 = 50 g (1)

ECF from an incorrect subtraction

Worked solution

(a) A 15-year-old girl is still growing quickly. Making new cells and tissues needs energy from respiration, so she needs more energy each day than an adult who has stopped growing.

(b) The pregnant woman needs energy for the growth of the fetus and placenta. She also has extra mass to carry around, so moving uses more energy.

(c) Mass of water = 25 g. Energy transferred = 25 × 4.2 × 12 = 1260 J. Energy per gram = 1260 ÷ 0.80 = 1575 J per g, which is about 1600 J per g.

(d) Extra energy each day = 9200 − 8400 = 800 kJ. Each gram of cereal provides 16 kJ, so mass = 800 ÷ 16 = 50 g. (Notice the packet value, 16 kJ per g, is about ten times the student's value; simple burning experiments lose a lot of energy to the surroundings.)

Topical Hub
topical-hub.pages.dev

17 · Part 2 Paper 1 & 2 Nutrition [7]

Mark scheme

(a)

(a)(i) **[1]**

- colon (1)

ALLOW large intestine

(a)(ii) **[1]**

- rectum (1)

REJECT anus

(a)(iii) **[1]**

- oesophagus (1)

ALLOW gullet

(b) **[2]**

- produced by the liver (1)
- stored in the gall bladder (1)

(c) **[2]**

- the pancreas produces OR secretes digestive enzymes, e.g. amylase OR protease OR lipase (1)
- which are released into the small intestine OR duodenum (1)

Worked solution

(a) (i) Water is absorbed from the undigested food in the colon, part of the large intestine, so the faeces become more solid. (ii) Faeces are stored in the rectum before egestion through the anus. (iii) The oesophagus carries food from the mouth to the stomach.

(b) Bile is made in the liver and stored in the gall bladder until food enters the small intestine.

(c) The pancreas makes digestive enzymes, such as amylase, protease and lipase. These pass along a duct into the duodenum, where they digest starch, proteins and lipids.

Topical Hub
topical-hub.pages.dev

18 · Part 2 Paper 1 & 2 Nutrition [8]

Mark scheme

(a) **[2]**

- lipase digests OR breaks down lipid (fat) in the milk (1)
- into fatty acids (and glycerol), and the fatty acids lower the pH (1)

(b) **[3]**

- bile (salts) emulsify the lipid OR break lipid into small droplets (1)
- this gives a larger surface area (1)
- for lipase to act on, so lipid is digested faster OR fatty acids are produced faster (1)

REJECT bile digests lipid OR bile is an enzyme

(c) **[2]**

- boiling denatured the lipase (1)
- the active site changed shape, so lipid could not bind OR no fatty acids were produced, so the pH did not change (1)

REJECT the enzyme was killed

(d) **[1]**

- temperature affects the rate of enzyme activity, so it is a control variable OR so that only the presence of bile (or boiling the lipase) affects the results (1)

ALLOW to make it a fair test

Worked solution

(a) Lipase digests the lipid in milk into fatty acids and glycerol. Fatty acids are acidic, so the pH falls from 9.0 towards 7.0.

(b) Bile salts emulsify the lipid: they break large fat drops into many tiny droplets. The tiny droplets have a much larger total surface area, so lipase molecules can reach more lipid at once, and digestion is faster. Bile is not an enzyme; it does not digest the lipid itself.

(c) Boiling denatured the lipase. Its active site changed shape, so lipid could no longer fit, no fatty acids were made, and the pH stayed the same. The bile salts alone could not change the pH because bile does not digest lipid.

(d) Temperature affects how fast enzymes work. Keeping all tubes at 35 °C means any difference in the results is caused only by the substances added.

Topical Hub
topical-hub.pages.dev

19 · Part 2

Paper 1 & 2 Nutrition [8]

Mark scheme

(a) **[1]**

- lacteal (1)

ALLOW lymph vessel

(b) **[5]**

Award one mark for each point, up to a maximum of 5.

- villi (and microvilli) give a large surface area for absorption (1)
- the wall of a villus is one cell thick, so there is a short distance for diffusion (1)
- the dense network of capillaries carries absorbed food away (1)
- which maintains a steep concentration gradient (for diffusion) (1)
- the lacteal absorbs fatty acids and glycerol (1)
- the small intestine is very long, so there is a lot of time for absorption OR a large surface area (1)
- epithelial cells have many mitochondria to release energy for active transport (1)

(c) **[2]**

- flattened villi have a smaller surface area (1)
- so less digested food (e.g. glucose, amino acids) is absorbed, so the body uses its stored fat OR has less energy and materials for growth (1)

Worked solution

(a) The central vessel is the lacteal, which absorbs fatty acids and glycerol.

(b) The small intestine is long, and its wall is folded into millions of villi; the villi cells also have microvilli. Together these give a huge surface area. The villus wall is only one cell thick, so digested food has a short distance to travel. Each villus has a dense capillary network, and the blood constantly carries absorbed food away, so there is always a steep concentration gradient between the gut and the blood. Fatty acids and glycerol are absorbed into the lacteal. Some substances are absorbed by active transport, using energy released by the many mitochondria in the epithelial cells.

(c) Flattened villi have a much smaller surface area, so less digested food is absorbed even if the person eats well. The body then has to use its energy stores, such as fat, and cannot get enough materials for growth, so the person loses mass.

Topical Hub
topical-hub.pages.dev

20 · Part 2

Paper 2 only | Nutrition | [8]

Mark scheme

(a) **[1]**

- (27 + 29 + 31) ÷ 3 = 29 °C (1)

(b) **[2]**

- 4.9 ÷ 25 × 100 (1)
- 19.6 (%) OR 20 (%) (1)

(c) **[3]**

Award one mark for each point, up to a maximum of 3.

- energy (heat) is transferred to the surrounding air instead of to the water (1)
- energy is used to heat the boiling tube OR the thermometer OR the clamp or needle (1)
- the peanut may not burn completely OR some of the peanut is left unburned (1)
- energy is lost while the burning peanut is moved from the Bunsen burner to the boiling tube (1)
- the flame is not always directly under the boiling tube (1)

(d) **[2]**

Award one mark for each improvement, up to a maximum of 2.

- insulate the boiling tube OR use a calorimeter OR put a draught shield around the apparatus (1)
- relight the peanut until it no longer burns OR until it is fully burned (1)
- hold the peanut closer to the boiling tube, or keep it directly under the tube (1)
- stir the water before reading the highest temperature (1)

ALLOW use a metal can instead of a glass boiling tube

Worked solution

(a) Mean = (27 + 29 + 31) ÷ 3 = 87 ÷ 3 = 29 °C.

(b) Percentage = (4.9 ÷ 25) × 100 = 19.6%, about 20%. Amira's value is only about one fifth of the packet value.

(c) Much of the energy released by the burning peanut never reaches the water. It heats the air around the apparatus, the boiling tube and the thermometer. Also, the peanut may not burn completely, so not all of its energy is released. Some energy is lost while the lit peanut is moved from the Bunsen burner to the boiling tube.

(d) Insulating the boiling tube or using a calorimeter reduces energy losses to the air. Relighting the peanut until it will no longer burn makes sure all its energy is released. Keeping the peanut close to and directly under the tube, and stirring the water, also help the reading be more accurate.

Topical Hub
topical-hub-pages.dev